

$$\boxed{A1} \begin{cases} \alpha x + 2y - 2z = 2 \\ 2x + 2y - 2z = \alpha \\ \alpha x + 2y - z = 1 \end{cases}$$

Boatzi $\alpha = 1$ daueau abt bod.

$$A' = \left(\begin{array}{ccc|c} \alpha & 2 & -2 & 2 \\ 2 & 2 & -2 & \alpha \\ \alpha & 2 & -1 & 1 \end{array} \right)$$

$\underbrace{\hspace{10em}}_A$

$$|A| = \begin{vmatrix} \alpha & 2 & -2 \\ 2 & 2 & -2 \\ \alpha & 2 & -1 \end{vmatrix} =$$

$$= -2\alpha - \cancel{y\alpha} - 8 + 4\alpha + 4 + \cancel{y\alpha} \\ = 2\alpha - 4$$

$$|A| = 2\alpha - 4$$

$$|A| = 0 \rightarrow 2\alpha - 4 = 0 \quad \alpha = 2$$

Kosuk:

• $\alpha \neq 2$ daueau $\text{heiu}(A) = \text{heiu}(A') = 3 \rightarrow$

SIST.
BATERAJANI
DETERMINANA
(Rouche Two)

• $\alpha = 2$ daueau $\rightarrow |A| = 0$

$$A' = \left(\begin{array}{ccc|c} 2 & 2 & -2 & 2 \\ 2 & 2 & -2 & 2 \\ 2 & 2 & -1 & 1 \end{array} \right)$$

Aren heio atterheko kantutor
boten da z1 eta z2 baraiwak
direta.

$$A \Rightarrow \begin{vmatrix} 2 & -2 \\ 2 & -1 \end{vmatrix} = -2 + 4 = 2 \neq 0 \quad \left. \vphantom{\begin{vmatrix} 2 & -2 \\ 2 & -1 \end{vmatrix}} \right\} \text{heiu}(A) = 2$$

$|A| = 0$

$$A' \quad \begin{vmatrix} 2 & -2 & 2 \\ 2 & -2 & 2 \\ 2 & -1 & 1 \end{vmatrix} = \cancel{-4} - \cancel{4} - \cancel{4} + \cancel{4} + \cancel{4} + \cancel{4} = 0 \quad \text{heiu}(A') = 2$$

$\text{heiu}(A) = \text{heiu}(A') = 2 \rightarrow$ SIST BATERAJ. INDETERMINANA

EMATZA

$\alpha \neq 2 \rightarrow$ SIST. BATERAJANI DETERMINANA
 $\alpha = 2 \rightarrow$ SIST. BATERAJ. INDETERMINANA

(2)

 $\alpha = 1$ daueau

$$\text{rank}(A) = \text{rank}(A') = 3 = n$$

SIST. BATERAPAN:

DETERMINAN $\rightarrow$ CRAMER

$$A' = \left(\begin{array}{ccc|c} 1 & 2 & -2 & 2 \\ 2 & 2 & -2 & 1 \\ 1 & 2 & -1 & 1 \end{array} \right)$$

$$|A| = -2 - 4 - 8 + 4 + 4 + 4 = -2$$

$$x = \frac{|A_x|}{|A|} = \frac{\begin{vmatrix} 2 & 2 & -2 \\ 1 & 2 & -2 \\ 1 & 2 & -1 \end{vmatrix}}{-2} = \frac{2}{-2} = \boxed{-1}$$

$$y = \frac{|A_y|}{|A|} = \frac{\begin{vmatrix} 1 & 2 & -2 \\ 2 & 1 & -2 \\ 1 & 1 & -1 \end{vmatrix}}{-2} = \frac{1}{2} \boxed{}$$

$$z = \frac{|A_z|}{|A|} = \frac{\begin{vmatrix} 1 & 2 & 2 \\ 2 & 2 & 1 \\ 1 & 2 & 1 \end{vmatrix}}{-2} = \frac{2}{-2} = \boxed{-1}$$

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A1

$$\begin{cases} x + y + \alpha z = \alpha \\ 2x + \alpha y + \alpha z = 1 \\ x + \alpha y + z = 1 \end{cases}$$

- Etabaizatu α -ren araberaz.

- Ebatzi $\alpha = -1$ eta $\alpha = 1$ balizetarako ahol bade.

Sistema etabaizatzeko Rouchen teorema erabiliko gara, koefizienteen matizearen heina eta matrice zabaldueen heina atartuz.

Sistemak hiru ezezagun ditu $\rightarrow \boxed{n=3}$

$$A' = \left(\begin{array}{ccc|c} 1 & 1 & \alpha & \alpha \\ 2 & \alpha & \alpha & 1 \\ 1 & \alpha & 1 & 1 \end{array} \right)$$

$\underbrace{\hspace{10em}}_A$

A eta A' -ren heinak 3 edo txikiagoak izan daitezke.

$$|A| = \alpha + \alpha + 2\alpha^2 - \alpha^2 - 2\alpha^2 = 2\alpha - 2$$

$$|A| = 0 \text{ denean } 2\alpha - 2 = 0 \quad 2(\alpha - 1) = 0 \rightarrow \boxed{\alpha = 1}$$

Kasuak

$\alpha \neq 1$ denean $\rightarrow |A| \neq 0 \rightarrow \text{heina}(A) = 3$.

$\text{heina}(A) = \text{heina}(A') = 3 = n$ SIST. BATERAJARRI DETERMINATUA.

$\alpha = 1$ denean $\rightarrow |A| = 0 \rightarrow \text{heina}(A) \leq 2$

$$A \rightarrow \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} = 1 - 2 = -1 \neq 0 \quad \boxed{\text{heina}(A) = 2}$$

$$A' \rightarrow \begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} = 0 \quad \text{heina}(A') = 2$$

$\rightarrow$ lerro berdineak.

$\left. \begin{array}{l} \text{heina}(A) = \text{heina}(A') \\ n = 3 \end{array} \right\}$ SIST. BATERAJARRI INDETERMINATUA.

$\alpha = -1$ daucau $\rightarrow$

SIST. BATERAFARRI
DETERMINATA daucau
CRAMERREKIN ebatiko da:

$$A' = \begin{pmatrix} 1 & 1 & -1 & -1 \\ 2 & -1 & -1 & 1 \\ 1 & -1 & 1 & 1 \end{pmatrix}$$

A

$$|A| = -4$$

$$x = \frac{\begin{vmatrix} -1 & 1 & -1 \\ 1 & -1 & -1 \\ 1 & -1 & 1 \end{vmatrix}}{-4} = \frac{-1+1-1+1-1+1}{-4} = 0 //$$

$$y = \frac{\begin{vmatrix} 1 & -1 & -1 \\ 2 & 1 & -1 \\ 1 & 1 & 1 \end{vmatrix}}{-4} = \frac{1+1-2+1+2+1}{-4} = -1 //$$

$$z = \frac{\begin{vmatrix} 1 & 1 & -1 \\ 2 & -1 & 1 \\ 1 & -1 & 1 \end{vmatrix}}{-4} = \frac{-1+1+2-1-2+1}{-4} = 0 //$$

$\alpha = 1$ daucau $\rightarrow$ SIST. BATERAFARRI INDETERM.

$$A' = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

1. eta 3. errenkada
bataiatuak direla, hirugarren
ekuazioa eta dozu erabotuko.

$$\begin{cases} x + y = 1 - z \\ 2x + y = 1 - z \end{cases} \Rightarrow$$

$$\begin{cases} x + y = 1 - z \\ 2x + y = 1 - z \end{cases} \quad \begin{pmatrix} 1 & 1 & 1-z \\ 2 & 1 & 1-z \end{pmatrix}$$

B $|B| = -1$

$$z = 1.$$

$$x = \frac{\begin{vmatrix} 1-z & 1 \\ 1-z & 1 \end{vmatrix}}{-1} = 0.$$

$$y = \frac{\begin{vmatrix} 1 & 1-z \\ 2 & 1-z \end{vmatrix}}{-1}$$

$$y = \frac{1-z-2+2z}{-1} = -1+1$$

$$\boxed{x=0, y=1-1, z=1}$$

A1

$$\begin{cases} ax - y + z = 1 \\ 3x - y + az = a \\ x + (a-1)z = 1 \end{cases}$$

Rouché's theorem erabakitako dozu sistema nola a + tertuko

$$A' = \left(\begin{array}{ccc|c} a & -1 & 1 & 1 \\ 3 & -1 & a & a \\ 1 & 0 & a-1 & 1 \end{array} \right)$$

$\underbrace{\hspace{10em}}_A$

$$|A| = \begin{vmatrix} a & -1 & 1 \\ 3 & -1 & a \\ 1 & 0 & a-1 \end{vmatrix} =$$

$$= -a(a-1) - a + 1 + 3(a-1) =$$

$$= -a^2 + \cancel{a} - \cancel{a} + 1 + 3a - 3 =$$

$$= -a^2 + 3a - 2.$$

$$|A| = 0.$$

$$-a^2 + 3a - 2 = 0.$$

$$a^2 - 3a + 2 = 0 \quad \begin{cases} a = 2 \\ a = 1 \end{cases}$$

Kasuak

$a \neq 2$ eta $a \neq 1$ denean $\rightarrow |A| \neq 0 \rightarrow \text{rang}(A) = \text{rang}(A') = 3$

SIST. BATERA FARRI
DETERMINATA

$a = 2$ denean.

$$A \rightarrow \left. \begin{array}{l} |A| = 0. \\ \begin{vmatrix} 3 & -1 \\ 1 & 0 \end{vmatrix} \neq 0 \end{array} \right\} \text{rang}(A) = 2$$

$a = 2$ denean:

$$A' = \left(\begin{array}{ccc|c} 2 & -1 & 1 & 1 \\ 3 & -1 & 2 & 2 \\ 1 & 0 & 1 & 1 \end{array} \right)$$

$$A' \rightarrow \begin{vmatrix} 2 & -1 & 1 \\ 3 & -1 & 2 \\ 1 & 0 & 1 \end{vmatrix} = 0. \text{ rang}(A') = 2.$$

SIST. BATERA FARRI INDETER.

$a=1$ deueau. $\rightarrow |A|=0$

$$A' = \left(\begin{array}{cc|cc} 1 & -1 & 1 & 1 \\ 3 & -1 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{array} \right) \quad \left. \begin{array}{l} \begin{vmatrix} 1 & -1 \\ 3 & -1 \end{vmatrix} \neq 0 \\ |A|=0 \end{array} \right\} \text{heima } (A)=2$$

$$A' \rightarrow \left(\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 3 & -1 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{array} \right) = 2 \rightarrow \text{heima } (A')=3$$

SIST. BATERMEKINS

$a \neq 2$ eto $a \neq 1$	SIST. BATERJARKI DETERM.
$a=2$	SIST. BATERJARKI INDET.
$a=1$	SIST. BATERMEKINS

Etbaži sistema $a=3$ deueau:

$$A' = \left(\begin{array}{ccc|c} 3 & -1 & 1 & 1 \\ 3 & -1 & 3 & 3 \\ 1 & 0 & 2 & 1 \end{array} \right)$$

A

$a=3$ deueau sist BATERJ
DETERMINATA daret
CRAMER-reka etatulo doju.
 $|A|=-2$.

$$x = \frac{|A_x|}{|A|} = \frac{\begin{vmatrix} 1 & -1 & 1 \\ 3 & -1 & 3 \\ 1 & 0 & 2 \end{vmatrix}}{-2} = \frac{2}{-2} = -1$$

$$y = \frac{|A_y|}{|A|} = \frac{\begin{vmatrix} 3 & 1 & 1 \\ 3 & 3 & 3 \\ 1 & 1 & 2 \end{vmatrix}}{-2} = \frac{6}{-2} = -3$$

$$z = \frac{|A_z|}{|A|} = \frac{\begin{vmatrix} 3 & -1 & 1 \\ 3 & -1 & 3 \\ 1 & 0 & 1 \end{vmatrix}}{-2} = \frac{-2}{-2} = 1$$

Emirna

$$\boxed{\begin{array}{l} x=-1 \\ y=-3 \\ z=1 \end{array}}$$

EBAU - 2021 Et OHIKOZA.

A1 $\begin{cases} \alpha x + 2y - z = \alpha \\ 2x + \alpha y + z = 2 + \alpha \\ x - \alpha y + 2z = 2\alpha \end{cases}$ • Etabaidatu α -ren a/oben
• Ebati $\alpha = 1$ kasuak.

Etabaidatzeko A eta A' matritzen heinak atzeratu dira
Roucheren erregelaren bidez.

$$A' = \left(\begin{array}{ccc|c} \alpha & 2 & -1 & \alpha \\ 2 & \alpha & 1 & 2+\alpha \\ 1 & -\alpha & 2 & 2\alpha \end{array} \right)$$

$\underbrace{\hspace{10em}}_A$

$$|A| = 2\alpha^2 + 2 + 2\alpha + \alpha - 8 + \alpha^2 = 3\alpha^2 + 3\alpha - 6 = 0$$

$$\alpha = \frac{-3 \pm \sqrt{3^2 - 4 \cdot 3 \cdot (-6)}}{2 \cdot 3} = \begin{cases} -2 \\ 1 \end{cases}$$

Kasuak:

• $\alpha \neq 1$ eta $\alpha \neq -2$ dauean $\rightarrow |A| \neq 0 \rightarrow$ SISTEMA BATERAGARRI DETERMINATUA
 $\text{heiu}(A) = \text{heiu}(A') = 3 = n$

• $\alpha = -2$ dauean: $\rightarrow |A| = 0$

$$A' = \left(\begin{array}{ccc|c} -2 & 2 & -1 & -2 \\ 2 & -2 & 1 & 0 \\ 1 & 2 & 2 & -4 \end{array} \right)$$

$$\begin{vmatrix} 2 & 2 \\ 2 & -2 \end{vmatrix} = 0$$

$$\begin{vmatrix} 2 & -2 \\ 1 & 2 \end{vmatrix} = 6 \neq 0$$

$\left. \begin{array}{l} \text{heiu}(A) \neq \text{heiu}(A') \\ \text{heiu}(A) = 2 \end{array} \right\}$ BATERA-EGINA

$$A' \rightarrow \begin{vmatrix} -2 & 2 & -2 \\ 2 & -2 & 4 \\ 1 & 2 & -4 \end{vmatrix} = -16 + 16 - 4 + 16 + 16 = 12 \neq 0 \quad \left. \begin{array}{l} \text{heiu}(A') = 3 \end{array} \right\}$$

• $\alpha = 1$ dauean $\rightarrow |A| = 0$

$$A' = \left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 2 & 1 & 1 & 3 \\ 1 & -1 & 2 & 2 \end{array} \right)$$

$$A \rightarrow \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} = 1 - 4 = -3 \neq 0$$

$\text{heiu}(A) = 2$

$$A' \rightarrow \begin{vmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 2 \end{vmatrix} = 2 + 6 - 2 - 1 - 3 + 3 = 0 \quad \text{heiu}(A') = 2$$

SIST. BATERAGARRI INDETERMINATUA

$\text{heiu}(A) = \text{heiu}(A') = 2 < n$

Sistema batzori indeterminatua $\text{heiu}(A) = \text{heiu}(A') = 2$
 Cramer erabiltzeko $|A| \neq 0$ itau behar da eta horiek
 sistema batzori determinatua bihurtu behar da.
 Horretarako ezetapu bat parametro bihurtu
 dozu $\rightarrow z=1$ eta ekuazio bat erabiltzeko

$$\alpha=1 \rightarrow \begin{cases} x+2y-z=1 \\ 2x+y+z=3 \end{cases}$$

$$\begin{cases} x+2y=1+1 \\ 2x+y=3-1 \end{cases} \Rightarrow \underbrace{\begin{pmatrix} 1 & 2 & | & 1+1 \\ 2 & 1 & | & 3-1 \end{pmatrix}}_A = A' \quad |A| = -3$$

CRAMER

$$x = \frac{|A_x|}{|A|} = \frac{\begin{vmatrix} 1+1 & 2 \\ 3-1 & 1 \end{vmatrix}}{-3} = \frac{1+1-6+2}{-3} = \frac{-5+3}{-3}$$

$$y = \frac{|A_y|}{|A|} = \frac{\begin{vmatrix} 1 & 1+1 \\ 2 & 3-1 \end{vmatrix}}{|A|} = \frac{3-1-2-2}{-3} = \frac{-3+1}{-3}$$

$$z=1$$

EMAITZA:

$$\boxed{x = \frac{5-3}{3}, y = 1-\frac{1}{3}, z=1}$$



SAEKTIBITATEA EKAINA 2020.

A1

$$S(a) = \begin{cases} ax - y + 2z = 2 \\ x - 2y - z = 1 \\ x + 2y + az = 3 \end{cases}$$

$$A' = \left(\begin{array}{ccc|c} a & -1 & 2 & 2 \\ 1 & -2 & -1 & 1 \\ 1 & 2 & a & 3 \end{array} \right)$$

$$|A| = -2a^2 + 1 + 4 + 4 + a + 2a = -2a^2 + 3a + 9$$

$$|A| = 0 \quad 2a^2 - 3a - 9 = 0 \quad a = \frac{3 \pm \sqrt{3^2 - 4 \cdot 2 \cdot (-9)}}{2 \cdot 2} = \frac{3 \pm 9}{4} = \begin{matrix} 3 \\ -\frac{3}{2} \end{matrix}$$

Kasuak:

• $a \neq 3$ eta $a \neq -3/2$ dauean:

$$\text{heiu}(A) = \text{heiu}(A') = 3 = n$$

SIST. BATERA ERRI DETERMINATUA

• $a = 3$ dauean.

$$\left(\begin{array}{ccc|c} 3 & -1 & 2 & 2 \\ 1 & -2 & -1 & 1 \\ 1 & 2 & 3 & 3 \end{array} \right)$$

$$A \Rightarrow \begin{vmatrix} 3 & -1 \\ 1 & -2 \end{vmatrix} = -6 + 1 = -5 \quad \left. \begin{matrix} \text{heiu}(A) = 2 \\ |A| = 0 \end{matrix} \right\}$$

$\text{heiu}(A) \neq \text{heiu}(A')$
BATERA ERRI

$$A' \rightarrow \begin{vmatrix} 3 & -1 & 2 \\ 1 & -2 & 1 \\ 1 & 2 & 3 \end{vmatrix} = -14 \neq 0 \quad \text{heiu}(A') = 3$$

• $a = -3/2$ $\rightarrow |A| = 0$

$$\left(\begin{array}{ccc|c} -3/2 & -1 & 2 & 2 \\ 1 & -2 & -1 & 1 \\ 1 & 2 & -3/2 & 3 \end{array} \right)$$

$$A \rightarrow \begin{vmatrix} -3/2 & -1 \\ 1 & -2 \end{vmatrix} = 3 + 1 = 4 \neq 0$$

$$\text{heiu}(A) = 2.$$

$$A' \rightarrow \begin{vmatrix} -3/2 & -1 & 2 \\ 1 & -2 & 1 \\ 1 & 2 & 3 \end{vmatrix} = 22 \neq 0$$

$\text{heiu}(A) \neq \text{heiu}(A')$ $\leftarrow$
SIST. BATERA ERRI

Kalkulatu eiten dogu CRAMERKEE bidez
 $a \neq 3$ eta $a \neq -3/2$ denetan a parametroaren
arabera

$$|A| = -2a^2 + 3a + 9 = -2(a-3)(a+3/2)$$

$$x = \frac{\begin{vmatrix} 2 & -1 & 2 \\ 1 & -2 & -1 \\ 3 & 2 & a \end{vmatrix}}{-2(a-3)(a+3/2)} = \frac{-4a+3+4+12+a+4}{-2(a-3)(a+3/2)} = \frac{-3a+23}{-2(a-3)(a+3/2)}$$

$$y = \frac{\begin{vmatrix} a & 2 & 2 \\ 1 & 1 & -1 \\ 1 & 3 & a \end{vmatrix}}{-2(a-3)(a+3/2)} = \frac{a^2-2+6-2-2a+3a}{-2(a-3)(a+3/2)} = \frac{a^2+a+2}{-2(a-3)(a+3/2)}$$

$$z = \frac{\begin{vmatrix} a & -1 & 2 \\ 1 & -2 & 1 \\ 1 & 2 & 3 \end{vmatrix}}{-2(a-3)(a+3/2)} = \frac{-6a+1+4+4+3-2a}{-2(a-3)(a+3/2)} = \frac{-8a+12}{-2(a-3)(a+3/2)}$$

A1

$$\begin{cases} x + y + z = 2A \\ 2x + 3y + 4z = 2 \\ 4x + 4y + Az = 4A \end{cases}$$

Ebatzi posiblec dauenau.

$$B' = \left(\begin{array}{ccc|c} 1 & 1 & 1 & 2A \\ 2 & 3 & 4 & 2 \\ 4 & 4 & A & 4A \end{array} \right)$$

$\underbrace{\hspace{10em}}_B$

$$|B| = 3A + 16 + 8 - 12 - 2A - 16 = A - 4$$

$$|B| = 0 \quad A - 4 = 0 \rightarrow A = 4.$$

kasuak:

• $A \neq 4$ dauenau. $|B| \neq 0$

$$\text{heiu}(B) = \text{heiu}(B') = 3 = n$$

SIST. BATERAGARRI: DETERMINATUA• $A = 4$ dauenau. $|B| = 0$.

$$B \rightarrow \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix} \neq 0 \rightarrow \text{heiu}(B) = 2.$$

$$B' \rightarrow \begin{vmatrix} 1 & 1 & 8 \\ 2 & 3 & 2 \\ 4 & 4 & 16 \end{vmatrix} = -16 \neq 0 \quad \text{heiu}(B') = 3$$

 $\left. \begin{array}{l} \text{heiu}(B) \neq \text{heiu}(B') \\ \text{BATERAGARRI} \end{array} \right\}$
Ebatzi egin da $A \neq 4$ denetako. SIST. BAT. DETERM

CRAMEREN BATERAKETZ:

$$\begin{aligned} \boxed{X} = \frac{|B_x|}{|B|} &= \frac{\begin{vmatrix} 2A & 1 & 1 \\ 2 & 3 & 4 \\ 4A & 4 & A \end{vmatrix}}{A-4} = \frac{6A^2 + 16A + 8 - 12A - 2A - 32A}{A-4} \\ &= \boxed{\frac{6A^2 - 30A + 8}{A-4}} \end{aligned}$$

$$y = \frac{|B_y|}{|B|} = \frac{\begin{vmatrix} 1 & 2A & 1 \\ 2 & 2 & 4 \\ 4 & 4A & A \end{vmatrix}}{A-4} = \frac{2A + 32A + 8A - 8 - 4A^2 - 16A}{A-4}$$

$$= \frac{-4A^2 + 26A - 8A}{A-4}$$

$$z = \frac{|B_z|}{|B|} = \frac{\begin{vmatrix} 1 & 1 & 2A \\ 2 & 3 & 2 \\ 4 & 4 & 4A \end{vmatrix}}{A-4} =$$

$$= \frac{12A + \cancel{8} + 16A - 24A - \cancel{8} - 8A}{A-4} = \frac{-4A}{A-4}$$

$$\begin{cases} x + 2y + 3z = 6 \\ x + y - z = 1 \\ 2x - 2y + Az = A \end{cases}$$

Matritzabildung: $A' = \underbrace{\begin{pmatrix} 1 & 2 & 3 \\ 1 & 1 & -1 \\ 2 & -2 & A \end{pmatrix}}_{(A)} \begin{vmatrix} 6 \\ 1 \\ A \end{vmatrix}$

$$|A| = \begin{vmatrix} 1 & 2 & 3 \\ 1 & 1 & -1 \\ 2 & -2 & A \end{vmatrix} = A - 4 - 6 - 6 - 2A - 2 = -A - 18$$

$$|A| = 0 \rightarrow -A - 18 = 0 \rightarrow A = -18.$$

Kasual:

- $A \neq -18$ daueau $|A| \neq 0 \rightarrow \text{heiu}(A) = \text{heiu}(A') = 3 = n$
SIST. BATERAFARRI DETERMIN.

- $A = -18$ daueau $|A| = 0 \rightarrow \text{heiu}(A) \leq 2$.

$$A \Rightarrow \left. \begin{array}{l} \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} = -1 \neq 0 \\ |A| = 0 \end{array} \right\} \text{heiu}(A) = 2$$

$$A' \Rightarrow \begin{vmatrix} 1 & 2 & 6 \\ 1 & 1 & 1 \\ 2 & -2 & -18 \end{vmatrix} = -18 + 4 - 12 - 12 + 36 + 2 = 0. \text{heiu}(A') = 2$$

$$\text{heiu}(A) = \text{heiu}(A') = 2 < n$$

SIST. BATERAFARRI INDETERM.

EBAU 2019 - EKAINA

$$\begin{array}{l} \boxed{A1} \left\{ \begin{array}{l} (m+3)x + my + mz = m-1 \\ 3x + mz = m-2 \\ -y + z = m-3 \end{array} \right. \end{array}$$

Erabazi indeterminatua den kasuetan baldin eta hainbat kasu.

$$A' = \left(\begin{array}{ccc|c} m+3 & m & m & m-1 \\ 3 & 0 & m & m-2 \\ 0 & -1 & 1 & m-3 \end{array} \right)$$

$\underbrace{\hspace{10em}}_A$

$$|A| = -3m - 3m + m(m+3) = -6m + m^2 + 3m = m^2 - 3m$$

$$|A| = 0 \rightarrow m^2 - 3m = 0 \quad \begin{array}{l} m_1 = 0 \\ m_2 = 3 \end{array}$$

Kasuak:

- $m \neq 0$ eta $m \neq 3$ dauean
 $\text{heiu}(A) = 3 = \text{heiu}(A') = n$

SIST. BATERAFARRI
DETERMINATUA

- $m = 0$ dauean $\rightarrow |A| = 0$.

$$\left(\begin{array}{ccc|c} 3 & 0 & 0 & -1 \\ 3 & 0 & 0 & -2 \\ 0 & -1 & 1 & -3 \end{array} \right)$$

$$A \Rightarrow \left. \begin{array}{l} \begin{vmatrix} 3 & 0 \\ 0 & -1 \end{vmatrix} = -3 \neq 0 \\ |A| = 0 \end{array} \right\} \text{heiu}(A) = 2$$

$\text{heiu}(A) \neq \text{heiu}(A')$
BATERAFERRA

$$A' \Rightarrow \left(\begin{array}{ccc|c} 3 & 0 & 0 & -1 \\ 3 & 0 & 0 & -2 \\ 0 & -1 & 1 & -3 \end{array} \right) \quad \begin{vmatrix} 3 & 0 & -1 \\ 3 & 0 & -2 \\ 0 & -1 & -3 \end{vmatrix} = 3 - 6 = -3 \neq 0 \quad \text{heiu}(A') = 3$$

• $n = 3$ daueau $|A| = 0$

$$\left(\begin{array}{cc|cc} 6 & 3 & 3 & 2 \\ 3 & 0 & 3 & 1 \\ 0 & -1 & 1 & 0 \end{array} \right)$$

$$A \Rightarrow \begin{vmatrix} 6 & 3 \\ 3 & 0 \end{vmatrix} = -9 \neq 0 \quad \text{rank}(A) = 2$$

$$A' \Rightarrow \begin{vmatrix} 6 & 3 & 2 \\ 3 & 0 & 1 \\ 0 & -1 & 0 \end{vmatrix} = -6 + 6 = 0 \quad \text{rank}(A') = 2$$

$\text{rank}(A) = \text{rank}(A') < n$
SIST. BAUF.
INDETERM.

Crameren errefebget ebaten da, uolt eto Cramer determinantekin erabli. Aldeko epuet eto 3 erenkode konpau utato

$$\boxed{z = \lambda}$$

$$\begin{cases} 6x + 3y = 2 - 3\lambda \\ 3x = 1 - 3\lambda \end{cases}$$

$$\Rightarrow \left(\begin{array}{cc|c} 6 & 3 & 2-3\lambda \\ 3 & 0 & 1-3\lambda \end{array} \right)$$

$$x = \frac{\begin{vmatrix} 2-3\lambda & 3 \\ 1-3\lambda & 0 \end{vmatrix}}{-9} = \frac{-3(1-3\lambda)}{-9} = \frac{1-3\lambda}{3}$$

$$y = \frac{\begin{vmatrix} 6 & 2-3\lambda \\ 3 & 1-3\lambda \end{vmatrix}}{-9} = \frac{\cancel{6} - 18\lambda - \cancel{6} + 9\lambda}{-9} = \frac{-9\lambda}{-9} = \lambda$$

$$\boxed{\begin{aligned} x &= \frac{1-3\lambda}{3} = -\lambda + 1/3 \\ y &= \lambda \\ z &= \lambda \end{aligned}}$$

SELEKTIIVITATEA 2018 - ULTALIA.

$$S(a) = \begin{cases} x + ay - z = 2 \\ 2x + y + az = 0 \\ 3x + (a+1)y - z = a-1 \end{cases}$$

Ettehaidatu
et ebatü $a=1$.
daueau.

$$A' = \left(\begin{array}{ccc|c} 1 & a & -1 & 2 \\ 2 & 1 & a & 0 \\ 3 & a+1 & -1 & a-1 \end{array} \right)$$

A eta A' matriseu hiuck
kalkulatuz eto Rouchera
teoremu erabiltz:

$$\begin{aligned} |A| &= -1 + 3a^2 - 2(a+1) + 3 + 2a - a(a+1) \\ &= -1 + 3a^2 - 2a - 2 + 3 + 2a - a^2 - a = \\ &= 2a^2 - a \end{aligned}$$

$$|A|=0 \rightarrow 2a^2 - a = 0 \quad a(2a-1)=0 \quad \begin{cases} a=0 \\ a=1/2 \end{cases}$$

Kasuak:

• $a \neq 0$ eta $a \neq 1/2$ daueau $|A| \neq 0 \rightarrow$

$$\text{heiu}(A) = 3 = \text{heiu}(A') = n \quad \text{SIST. BATERAFAKRI. DETEKTINATIBA.}$$

• $a=0$ daueau $\rightarrow |A|=0$

$$A' = \left(\begin{array}{ccc|c} 1 & 0 & -1 & 2 \\ 2 & 1 & 0 & 0 \\ 3 & 1 & -1 & -1 \end{array} \right)$$

$$A \rightarrow \begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} = 1 \neq 0 \quad \text{heiu}(A)=2$$

$$|A|=0$$

$\text{heiu}(A) \neq \text{heiu}(A')$
SIST. BATERDEZINA

$$A' \rightarrow \begin{vmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 3 & 1 & -1 \end{vmatrix} = -3 \neq 0 \quad \text{heiu}(A')=3$$

• $a = 1/2$ daueau $|A| = 0$

$$\left(\begin{array}{cc|c} 1 & 1/2 & -1 & 2 \\ 2 & 1 & 1/2 & 0 \\ 3 & 3/2 & -1 & -1/2 \end{array} \right)$$

$$A \rightarrow \left\{ \begin{array}{l} \begin{vmatrix} 1/2 & -1 \\ 1 & 1/2 \end{vmatrix} = \frac{1}{4} + 1 = \frac{5}{4} \neq 0 \\ |A| = 0 \end{array} \right\} \text{heiu}(A) = 2$$

$$A' \rightarrow \begin{vmatrix} 1/2 & -1 & 2 \\ 1 & 1/2 & 0 \\ 3/2 & -1 & -1/2 \end{vmatrix} = -\frac{33}{8} \neq 0 \quad \text{heiu}(A') = 3.$$

$\text{heiu}(A) \neq \text{heiu}(A') \rightarrow \text{BATEKAEZINA}$

$a=1 \quad \left(\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 2 & 1 & 1 & 0 \\ 3 & 2 & -1 & 0 \end{array} \right) \Rightarrow \text{SIST. BATERAFARNI}$
 DETERMINATV
 (Cramer)
 $|A| = 1.$

$$x = \frac{\begin{vmatrix} 2 & 1 & -1 \\ 0 & 1 & 1 \\ 0 & 2 & -1 \end{vmatrix}}{1} = -6$$

$$y = \frac{\begin{vmatrix} 1 & 2 & -1 \\ 2 & 0 & 1 \\ 3 & 0 & -1 \end{vmatrix}}{1} = 10$$

$$z = \frac{\begin{vmatrix} 1 & 1 & 2 \\ 2 & 1 & 0 \\ 3 & 2 & 0 \end{vmatrix}}{1} = 2$$

ETIATTA

$$\boxed{\begin{array}{l} x = -6 \\ y = 10 \\ z = 2 \end{array}}$$